

Lukas Gosch

18/5/1993 | Citizenship: Austrian

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My research lies at the intersection of Machine Learning (ML) and optimization. During my PhD, I have worked on graph neural networks, and on using mathematical programming to prove properties about neural networks such as robustness. Currently, my research focuses on ML for combinatorial optimization – in particular, using ML to speed up traditional mixed-integer programming solvers.

EDUCATION

PHD MACHINE LEARNING

TECHNICAL UNIVERSITY OF MUNICH

January 2022 - now

- Supervised by Prof. Stephan Günnemann
- Part of the Konrad Zuse School of Excellence in Reliable AI

MACHINE LEARNING SUMMER SCHOOL (MLSS)

March 2024 - Okinawa Institute of Science and Technology (OIST) & RIKEN AIP, Japan

MSC COMPUTATIONAL SCIENCE

UNIVERSITY OF VIENNA

March 2018 - July 2021

- Focus: Machine Learning and Optimization
- Master's thesis award in Operations Research
- GPA: 1.0 (1 best, 5 worst)

BSC TECHNICAL PHYSICS

TECHNICAL UNIVERSITY OF VIENNA

September 2013 - September 2017

- Thesis: C++ data analysis in high energy physics
- GPA: 1.9 (1 best, 5 worst)

ACADEMIC HONORS & AWARDS

BEST REVIEWER

ICML 2025

BEST PAPER AWARD

AdvML-Frontiers @ NeurIPS 2024

BEST MASTER'S THESIS PRICE

Awarded by the Austrian Society of Operations Research (ÖGOR), 2021

DEAN'S LIST AWARD

University of Vienna - Faculty of Physics, 2021

PERFORMANCE SCHOLARSHIP

University of Vienna, 2020

TEACHING

TECHNICAL UNIVERSITY OF MUNICH

MASTER THESES SUPERVISED

- Morgane Ayle, Francesco Campi, Daniel Sturm, Philipp Foth, Ajinkya Mohgaonkar

GUIDED RESEARCH SUPERVISED

- Xichuan Chen

COURSE INSTRUCTOR

- Machine Learning
- ML for Graphs & Sequential Data
- Seminar: Selected Topics in ML Research

WORK EXPERIENCES

AMAZON

APPLIED SCIENTIST INTERNSHIP, SUPPLY CHAIN OPTIMIZATION

September 2025 – Februar 2026 | Luxembourg

Improved large-scale middle-mile network design optimization via realistic network speed estimations.

- Statistical modelling and efficient algorithmic implementation (Rust), improving accuracy by >10%pt.
- Estimated business impact of \$20M+ in 2026 alone

TECHNICAL UNIVERSITY OF MUNICH

PHD STUDENT, DATA ANALYTICS AND MACHINE LEARNING (DAML) GROUP

January 2022 - now | Munich, Germany

Research Topics:

- Machine Learning for Combinatorial Optimization
- Graph Neural Networks
- Adversarial Robustness and Certification

Published at main ML conferences such as NeurIPS and ICLR. My works got featured for spotlight presentation and won a best paper award.

AIT AUSTRIAN INSTITUTE OF TECHNOLOGY

MASTER THESIS, ENERGY LOGISTICS & TRANSPORT OPTIMIZATION GROUP

July 2020 – June 2021 | Vienna, Austria

Combinatorial optimization applied in green logistics

- Mathematical modelling introducing a novel network design problem
- Exact and heuristic solving using integer linear programming and hybrid heuristics (C++)

INSTITUTE OF SCIENCE & TECHNOLOGY AUSTRIA (ISTA)

RESEARCH INTERNSHIP, MACHINE LEARNING AND COMPUTER VISION GROUP

October 2019 - April 2020 | Vienna, Austria

Supervised by Prof. Christoph Lampert | Technical Report

- Information-theoretic approaches for deep self-supervised representation learning

FRAUNHOFER INSTITUTE FOR ALGORITHMS & SCIENTIFIC COMPUTING

RESEARCH INTERNSHIP, NUMERICAL DATA DRIVEN PREDICTION RESEARCH GROUP

July 2018 – September 2018 | Bonn, Germany

Python - Topological data analysis applied to time series

QUEEN'S UNIVERSITY BELFAST

RESEARCH INTERNSHIP, SCHOOL OF PSYCHOLOGY

July 2017 – August 2017 | Northern Ireland, UK

Matlab - Ball trajectory reconstruction from 3D markers

2008 - 2018 | Further employments

PUBLICATIONS

* ... Equal Contribution

- A. Mohgaonkar, **L. Gosch**, M. Sabanayagam, D. Ghoshdastidar, S. Günnemann "Exact Certification of Neural Networks and Partition Aggregation Ensembles against Label Poisoning", Trustworthy AI @ ICLR 2026
- M. Sabanayagam*, **L. Gosch***, S. Günnemann, D. Ghoshdastidar "Exact Certification of (Graph) Neural Networks Against Label Poisoning" Spotlight @ ICLR 2025 - top 5.1%
- **L. Gosch***, M. Sabanayagam*, D. Ghoshdastidar, S. Günnemann "Provable Robustness of (Graph) Neural Networks Against Data Poisoning and Backdoor Attacks" Best Paper Award - AdvML @ NeurIPS 2024, Published TMLR 2025
- P. Foth*, **L. Gosch***, S. Geisler, L. Schwinn, S. Günnemann "Adversarial Robustness of Graph Transformers" TMLR 2025 - J2C Certification, Accepted for Presentation @ ICLR 2026
- M. Kollovich, **L. Gosch**, Y. Scholten, M. Lienen, S. Günnemann "Assesing Robustness via Score-based Adversarial Image Generation" TMLR 2024 - J2C Certification
- P. Foth, **L. Gosch**, S. Geisler, L. Schwinn, S. Günnemann "Relaxing Graph Transformers for Adversarial Attacks" Differentiable Almost Everything @ ICML 2024
- **L. Gosch***, S. Geisler*, D. Sturm*, B. Charpentier, D. Zügner, S. Günnemann "Adversarial Training for Graph Neural Networks: Pitfalls, Solutions, and New Directions" NeurIPS 2023
- F. Campi, **L. Gosch**, T. Wollschläger, Y. Scholten, S. Günnemann "Expressivity of Graph Neural Networks Through the Lens of Adversarial Robustness" AdvML @ ICML 2023
- **L. Gosch**, D. Sturm, S. Geisler, S. Günnemann "Revisiting Robustness in Graph Machine Learning" ICLR 2023, also TSRML (oral) & ML Safety @ NeurIPS 2022
- M. Ayle, J. Schuchardt, **L. Gosch**, D. Zügner, S. Günnemann "Training Differentially Private Graph Neural Networks with Random Walk Sampling", TSRML@NeurIPS 2022
- **L. Gosch**, M. Prandtstetter, K. F. Doerner "On Modelling and Solving Green Collaborative Transportation Planning" IPIC2021
- R. Feldbauer, **L. Gosch**, L. Lüftinger, P. Hyden, A. Flexer, T. Rattei "DeepNOG: fast and accurate protein orthologous group assignment" Bioinformatics 2020

PREPRINTS

- **L. Gosch***, X. Chen*, Y. Scholten*, S. Günnemann "Certifying Graph Neural Networks Against Label and Structure Poisoning", Under Review

FURTHER ACTIVITIES

INVITED TALKS

- VerifAI @ ICLR 2025 (Oral Presentation)
- INSAIT, Sofia, 2024
- Adv-ML @ NeurIPS 2024 (Oral Presentation)
- MDSI InterConnect, TUM, 2024
- Theoretical Foundations of AI Group, TUM, 2023
- TSRML @ NeurIPS 2022 (Contributed Talk)

REVIEWER

- ICML 2026, ICLR 2025
- ICML 2025 (Best Reviewer Award)
- NeurIPS 2025, ICLR 2024

STUDENT REPRESENTATIVE

KONRAD ZUSE SCHOOL OF EXCELLENCE IN
RELIABLE AI (GRADUATE SCHOOL)
2022 - 2023

BOOK CLUBS

DAML READING GROUP | 2022 - 2024

- Read "High-Dimensional Statistics: A Non-Asymptotic Viewpoint" by M. Wainwright
- Read "The Theory of Statistics and Its Applications" by D. Cox

CLASSIC LITERATURE & PHILOSOPHY | 2017 - 2021

- Founder - Book list

ATTENDING OF LECTURES AND EVENTS

ON PHILOSOPHY, ETHICS, AND MACHINE LEARNING
Examples

- Inquire - Ethic conversations at the LMU
- Philosophy courses - University of Vienna

SOFTWARE

DEEPNOG

Tool for protein similarity search using deep networks. Published in Bioinformatics.

SKILLS

- Python (PyTorch, PyG, NumPy, scikit-learn)
- Rust, Gurobi, $\text{\LaTeX}$ GIT, Linux
- Some Experiences in C, C++, CPLEX, Java, R, Matlab, SQL
- German (Native), English (C1, IELTS 8.5)

REFERENCES

References Upon Request.